

DHANUSH KRISHNA

Computer Science Graduate | Machine Learning & Cybersecurity

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PROFILE

Computer Science graduate of BITS Pilani Dubai, focused on machine learning and cybersecurity. Experienced in building LLM-based applications and RAG pipelines for environments that cannot use cloud AI, and in writing systems from scratch in Rust, WebAssembly, and WebGPU. Open to software, machine learning, and data roles — particularly where the interesting problem is the constraint rather than the model.

EXPERIENCE

Cybersecurity Intern — exida Jan 2026 – Jul 2026

- Performed cybersecurity work and developed a locally hosted LLM RAG pipeline to read and interpret P&ID (piping and instrumentation diagram) engineering drawings.

Tech Intern — Züblin / Strabag SE Jul 2025 – Aug 2025

- Worked with virtual machines and computer networks, supporting the company's IT infrastructure.

Intern — Standard Global Quality Services Jun 2024 – Aug 2024

- Handled data entry, organization, and reporting in Microsoft Excel to support quality-service operations.

PROJECTS

Loom — LLM Inference Engine in Rust and WebAssembly

- Runs a language model entirely in a browser tab with no backend and no ML libraries — the GGUF parser, quantised matmul kernels, attention, and tokeniser are all written from scratch in Rust and compiled to WebAssembly.
- 18 tokens/second in-browser, byte-identical output to the native build, and agreement with PyTorch layer by layer; the 469 MB model is cached in IndexedDB so a revisit loads in 0.6 s.

Escapement — Raft Consensus Under Deterministic Simulation

- A Raft implementation whose nodes never touch a clock, a socket, or a random number — the simulator owns all three, so an entire run is a pure function of its seed and any failure replays exactly.
- Includes a linearizability checker and a fuzz harness with repro minimisation; every bug it found is written up with the seed that reproduces it.

Orrery — SQL Query Engine in Rust and WebAssembly

- An analytical SQL engine with zero dependencies in the engine crate — lexer, parser, binder, cost-based optimizer, vectorized operators, and a Parquet reader, all executing in the page over 7.4M rows.
- Every stage of the query is inspectable, including each optimizer rewrite; faster than sql.js on 13 of the 15 queries measured, by as much as 128×.

Lucida — Physically-Based Path Tracer in WebGPU

- Monte Carlo light transport written in WebGPU compute shaders — intersection routines, a binned-SAH BVH, and a GGX microfacet BSDF, with multiple importance sampling and a wavefront architecture.
- Validated against an independent CPU reference tracer written in Rust; convergence is measured against sixteen independent renders rather than modelled.

Nexus — Private, Self-Hosted AI Workbench

- Fully self-hosted AI assistant for environments that cannot send data to cloud AI (regulated industries, air-gapped networks), built on Open WebUI with Ollama serving open-weight models and ChromaDB as the knowledge base.
- Built a custom smart-routing layer that inspects each request and dispatches it to the right specialist model (fast, reasoning, coding, or vision), plus document chat over uploaded files via RAG.

- Implemented a sandboxed code interpreter for document generation — static denylist, resource limits, and bubblewrap isolation — and a Blueprint Analyzer that reads instrument tags off P&ID diagrams with a vision model and cross-references a failure-mode knowledge base.

VoiceGuide AI — Multilingual Travel Itinerary Planner

- Voice-driven travel planner built with Python and Streamlit that generates detailed 8–12 activity itineraries from spoken input.
- Integrated Azure Cognitive Services (speech-to-text and text-to-speech) and Azure OpenAI to support 15+ languages with auto-detection and bilingual output, plus PDF export of the final itinerary.

Also built

- A FastAPI URL shortener with live click analytics, a Streamlit market dashboard on yfinance, a scheduled price watcher, and a PDF summarizer on the OpenAI API.

SKILLS

Programming: Python, Rust, TypeScript, Java, C, SQL, JavaScript, HTML/CSS, Bash

AI & Data: RAG, Ollama, ChromaDB, open-weight LLMs, OpenAI API, Azure OpenAI, Azure Cognitive Services, Pandas, Plotly, MATLAB

Frameworks & Tools: WebAssembly, WebGPU / WGSL, Streamlit, FastAPI, Open WebUI, SQLite, BeautifulSoup, Linux, Git / GitHub, Microsoft Office, Canva

EDUCATION

B.E. Computer Science — BITS Pilani, Dubai Campus 2022 – 2026

St. Antony's Public School 2020 – 2022

Abu Dhabi Indian School 2014 – 2020

CERTIFICATIONS

Eight certifications in AI, 2026. Microsoft — Transform your business with AI · Scale AI in your organization · Embrace responsible AI principles and practices · Create business value with AI · Leverage AI tools and resources for your business · AWS — Planning a Generative AI Project · Introduction to Generative AI: Art of the Possible · Google Cloud — Introduction to Generative AI

LANGUAGES

English (fluent) · Malayalam (fluent) · Hindi · Spanish (A1) · French (A1)

ACHIEVEMENTS

Repeat school topper in IBT, IOF Olympiad, ASSET, and Shastra Pratibha · Marrs Spelling Bee Finalist